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Systems and methods for group messaging using blockchain-based secure key exchange with key escrow fallback

Abstract

A system described herein may provide for secure group messaging between multiple participant devices, in which two or more of the participant devices support blockchain-based techniques for the secure sharing of keys used to secure the group messaging, and in which one or more of the participant devices (e.g., a third device) do not support such techniques. A key escrow system may maintain keys associated with the third device, and the other devices of the group messaging system may retrieve such keys from the key escrow system. Such other devices may attempt to obtain keys associated with the third device from a blockchain and may retrieve such keys from the key escrow system when the retrieval from the blockchain is unsuccessful. The other devices may additionally share their respective keys to the blockchain and the key escrow system, or the key escrow system may “pull” such keys from the blockchain.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This Application is a Continuation-in-Part of U.S. patent application Ser. No. 17/321,378 filed on May 14, 2021, titled “SYSTEMS AND METHODS FOR BLOCKCHAIN-BASED SECURE KEY EXCHANGE,” the contents of which are herein incorporated by reference in their entirety.

BACKGROUND

(1) Some encryption techniques, such as Public Key Infrastructure (“PKI”) techniques, may make use of public keys, which may be used to encrypt messages that may be decrypted using an associated private key, and/or may be used in other techniques. Public keys may be distributed or stored by key escrow systems, which may provide a single point of failure and/or an opportunity for a “man-in-the-middle” attack, via which public key may be obtained, modified, and/or otherwise compromised.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example overview of one or more embodiments described herein;
- (2) FIG. 2 illustrates an example of using a shared public key via a private blockchain system, in accordance with some embodiments, in a communication that utilizes double ratchet techniques to secure communications;
- (3) FIG. 3 illustrates an example of establishing a private blockchain system, in accordance with one or more embodiments;
- (4) FIG. 4 illustrates an example of using a shared public key via a private blockchain system, in accordance with some embodiments, in a communication that utilizes double ratchet techniques to secure communications;
- (5) FIGS. 5 and 6 illustrate example processes for using a shared public key via a private blockchain system, in accordance with some embodiments, in a communication that utilizes double ratchet techniques to secure communications;
- (6) FIG. 7 illustrates an example initiation of a group communication session, in accordance with some embodiments;
- (7) FIG. 8 illustrates an example of secure group messaging using a private blockchain system, in accordance with some embodiments;
- (8) FIG. 9 illustrates an example initiation of a group communication session with one or more participants that do not support blockchain-based key exchange techniques, in accordance with some embodiments;
- (9) FIG. 10 illustrates an example of obtaining public keys associated with active participants of a group communication session from a private blockchain system as well as from a Key Escrow System (“KES”), in accordance with some embodiments;

- (10) FIG. 11 illustrates an example of outputting an encrypted message package to multiple participants of a group communication session, in accordance with some embodiments;
- (11) FIG. 12 illustrates an example of recipients of an encrypted message package obtaining a public key of a sender of the encrypted message from a blockchain system and/or from a KES, in accordance with some embodiments;
- (12) FIGS. 13 and 14 illustrate an example of a sender of a message, which does not support blockchain-based key exchange techniques, obtaining a public key of one or more participants of a group communication session that is associated with blockchain-based key exchange techniques, in accordance with some embodiments;
- (13) FIG. 15 illustrates an example of outputting an encrypted message package to multiple participants of a group communication session, in accordance with some embodiments;
- (14) FIG. 16 illustrates an example of obtaining a public key, associated with a sender of an encrypted message via a group communication session that is associated with blockchain-based key exchange techniques, from a KES, in accordance with some embodiments;
- (15) FIG. 17 illustrates an example of a KES that is a node and/or viewer of a private blockchain system associated with a group communication session, in accordance with some embodiments;
- (16) FIG. 18 illustrates an example environment in which one or more embodiments, described herein, may be implemented;
- (17) FIG. 19 illustrates an example arrangement of a radio access network (“RAN”), in accordance with some embodiments; and
- (18) FIG. 20 illustrates example components of one or more devices, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (19) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.
- (20) Embodiments described herein provide for the secure maintaining and providing of information, such as public keys used in PKI techniques or other techniques, using a secure distributed ledger (e.g., “blockchain”) system. For example, embodiments described herein may utilize a blockchain system in lieu of a key escrow system in the exchange and/or providing of public keys in a Diffie-Hellman key exchange technique or other type of technique in which public keys are provided from one entity to another. For example, as discussed herein, a first entity may generate an asymmetric key pair that includes a public key and a private key, and may provide the public key to a blockchain system for retrieval by one or more other entities. For example, the entities may be engaged in a secure messaging session, in which messages are encrypted and may be decrypted using one or more keys, including the public key. For example, the messaging session may be associated with multiple key derivation functions, in which a first key or set of keys (e.g., “root” keys) are used in the generation of one or more other keys (e.g., “send” keys and/or “receive” keys). A root key derivation function, for example, may be performed by first and second entities engaged in the communication session to generate respective send and/or receive keys for each entity. The entities may make use of different private keys in their respective root derivation functions, but may utilize one or more shared public keys for their respective root derivation functions. Examples of techniques that make use of multiple key derivation functions, including a root derivation function utilizing one or more shared public keys, include a “double ratchet” encryption technique, a “Signal Protocol,” and/or other types of techniques.
- (21) In some embodiments, the contents of the secured ledger system (e.g., blockchain) may be publicly available or accessible, but the information stored therein may have no meaning or use to an attacker or other malicious user, in the context of obtaining public keys associated with PKI techniques or other techniques. In some embodiments, a private blockchain may be used, in which only authorized entities are able to access the information stored in the private blockchain. Further, as records stored in a blockchain system are immutable, an attacker may be prevented from

modifying public keys and thereby potentially having access to information encrypted based on such keys. For simplicity, embodiments described herein are described in the context of the secured ledger system being a blockchain system. However, similar concepts may apply to another type of secured ledger system other than a blockchain system.

(22) As shown in FIG. 1, a first entity, such as User Equipment (“UE”) **101-1**, may generate (at **102**) a set of keys. The set of keys may be, and/or may include, asymmetric key pair **103**, which may include private key **105** and public key **107**. Private key **105** may be used to decrypt information encrypted by public key **107**, may be used in a double ratchet and/or Signal Protocol technique, and/or may be used for other suitable purposes. UE **101-1** may securely store private key **105**, such as in a local storage device associated with UE **101-1**, in a private key store, in a Universal Integrated Circuit Card (“UICC”), and/or in some other secure location.

(23) As further shown, UE **101-1** may record (at **104**) the public key to a blockchain system, such as private blockchain system **109**. As noted above, private blockchain system **109** may be “private,” in that only authorized entities (e.g., authorized UEs, authorized users, etc.) may have access to the information stored in private blockchain system **109**. The information stored in private blockchain system **109** may be organized as “blocks,” where one particular record corresponds to one particular block, and/or where one block includes multiple records. A block may also include a reference to one or more other blocks (e.g., a “previous” block and/or a “next” block), such that a group of blocks and their associated references may form a “chain” of blocks. Further, private blockchain system **109** may include a set of nodes **111** that form a consensus regarding the blocks of which the blockchain is comprised, thus preventing unauthorized changes to the information stored in the blockchain, as well as resiliency in case any particular node **111** is compromised or otherwise becomes unavailable. In some embodiments, nodes **111** may each be implemented by a respective UE **101**, such as a mobile telephone, a workstation computer, a tablet, an Internet of Things (“IoT”) device, and/or other suitable type of device. While shown in the figure as being “external” to private blockchain system **109**, in practice, UE **101-1** may be, and/or may implement, a particular node **111** associated with private blockchain system **109**.

(24) When recording the public key to private blockchain system **109**, UE **101-1** may provide an identifier associated with UE **101-1** and/or other identifying information based on which the public key may be retrieved by an entity that may use the public key. For example, the record may include an Internet Protocol (“IP”) address associated with UE **101-1**, a device name, a user name, a Session Initiation Protocol (“SIP”) address, and/or some other suitable information associated with UE **101-1**.

(25) Additionally, or alternatively, the record may include a session identifier, which may identify a communication session between UE **101-1** and another UE, such as UE **101-2**. For example, UE **101-1** and UE **101-2** may be engaged in a secure messaging session, in which UE **101-1** and UE **101-2** exchange encrypted messages. The messaging session may be associated with a double ratchet technique, a Signal Protocol technique, and/or other type of technique in which UE **101-1** and UE **101-2** provide public keys to each other.

(26) Briefly, for example, for each message sent between UE **101-1** and UE **101-2**, a new public key may be generated (e.g., in an alternating fashion, where UE **101-1** generates a public key for a first message, UE **101-2** generates a public key for a second message, UE **101-1** generates a public key for a third message, and so on). As described below, these alternating shared public keys may serve as a first “ratchet” in a double ratchet technique to ensure security of the encrypted communications. For example, as discussed below, the public keys may be used to generate symmetric send and/or receive keys, which may be used to encrypt and/or decrypt communications between UE **101-1** and UE **101-2** without transmitting the send and/or receive keys between each other.

(27) Private blockchain system **109** is illustrated as including four nodes **111-1**, **111-2**, **111-3**, and **111-4**. In practice, private blockchain system **109** may include fewer nodes and/or additional nodes.

Private blockchain system **109** may, in some embodiments, be associated with one or more suitable authentication mechanisms whereby only authorized devices or systems are able to participate as nodes **111** of private blockchain system **109**. For example, as discussed herein, particular users or UEs **101** may be specified or automatically identified (e.g., based on an address book or contact list associated with a particular UE **101** that is authorized to access private blockchain system **109** and/or some other suitable technique). As such, unauthorized devices, such as UE **101-N**, may not be able to access information stored in private blockchain system **109** and/or may not be able to participate as a node **111** of private blockchain system **109**. Further, as the nodes **111** of private blockchain system **109** may only include trusted devices, the integrity of the information stored by private blockchain system **109** may be enhanced, as the likelihood of a compromise or attack of private blockchain system **109** may be reduced or eliminated. Further, the recording and/or accessing of private blockchain system **109** may be private with respect to external devices that do not have access to private blockchain system **109**.

(28) In some embodiments, separate private blockchain systems **109** may be established for discrete groups, such as messaging or conversation groups (e.g., where a first communication session includes a first set of participants and is associated with a first private blockchain system **109**, and where a second communication session includes a second set of participants and is associated with a second private blockchain system **109**). In some embodiments, a single private blockchain system **109** may be associated with multiple communication sessions, where keys associated with a given communication session may be protected by a set of smart contracts or other security mechanisms that are accessible only to participants in the communication session.

(29) Once recorded to private blockchain system **109**, public key **107** may be obtained (at **106**) by UE **101-2** and/or another device or system to which the information stored in private blockchain system **109** is available. For example, as noted above, UE **101-2** may be a UE with which UE **101-1** is engaged in a communication session via which UE **101-1** and UE **101-2** exchange encrypted communications. UE **101-2** may identify a record stored in private blockchain system **109** that includes the identifier associated with UE **101-1**, a communication session associated with UE **101-1** and UE **101-2**, and/or some other suitable identifier. In some embodiments, UE **101-2** may be a node **111** of private blockchain system **109**, and may maintain a full copy of the blockchain stored by private blockchain system **109**. In some embodiments, private blockchain system **109** may implement or be communicatively coupled to a search system, which may provide records that match a query. For example, UE **101-2** may query private blockchain system **109** for the identifier associated with UE **101-1** and/or a communication session with UE **101-1**, and private blockchain system **109** may return one or more blocks that include the identifier associated with UE **101-1** and/or the communication session. In some embodiments, private blockchain system **109** may return, and/or UE **101-2** may otherwise identify, the latest (e.g., most current) block matching the query. In this manner, in situations where UE **101-1** has provided multiple public keys to private blockchain system **109** over time, UE **101-2** may be able to retrieve and/or otherwise identify the newest public key. Identifying the newest public key may be useful in a double ratchet technique, a Signal Protocol technique, and/or some other technique in which an updated public key is used for each message between UE **101-1** and UE **101-2**. In some embodiments, public keys may be provided with sequence numbers or other information, based on which public keys associated with communications received in a non-sequential manner (e.g., out of order) may be identified.

(30) In some embodiments, the public key may be provided as part of a Diffie-Hellman exchange or some other suitable technique in which public keys associated with multiple UEs **101** are exchanged amongst each other. UE **101-2** may accordingly maintain and/or use (at **108**) the received public key for a double ratchet technique, a Signal Protocol technique, and/or in some other technique that involves a Diffie-Hellman key exchange or other type of key exchange procedure.

(31) FIG. 2 illustrates an example of using a shared public key (e.g., shared via private blockchain

system **109**, in accordance with some embodiments) in a communication technique that utilizes double ratchet techniques to secure communications between UE **101-1** and UE **101-2**. As shown, UE **101-1** and UE **101-2** may each include a respective messaging client **201**, which may send and/or receive encrypted communications. For example, messaging client **201-1**, associated with UE **101-1**, may encrypt communications based on techniques described herein, and may send the encrypted communications to messaging client **201-2** associated with UE **101-2**. Messaging client **201-2** may receive encrypted communications from UE **101-1** (e.g., from messaging client **201-1**), may decrypt the communications based on techniques described herein, and may present the decrypted communications (e.g., via a graphical user interface (“GUI”) or other suitable manner of presentation). Similarly, messaging client **201-2**, associated with UE **101-2**, may encrypt communications based on techniques described herein, and may send the encrypted communications to messaging client **201-1** associated with UE **101-1**. Messaging client **201-1** may receive encrypted communications from UE **101-2** (e.g., from messaging client **201-2**), may decrypt the communications based on techniques described herein, and may present the decrypted communications (e.g., via a GUI or other suitable manner of presentation).

(32) The encryption and/or decryption by messaging client **201-1** and/or messaging client **201-2** may be performed using a symmetric key that is generated based on a double ratchet technique, a Signal Protocol technique, and/or other suitable technique. For example, as discussed herein, the symmetric key may be used by a sender (e.g., UE **101-1**, in this example) to encrypt a communication, and may be used by a recipient (e.g., UE **101-2**, in this example) to decrypt the communication. The symmetric keys may be generated by respective Send Key Derivation Function (“KDF”) **205** and/or Receive KDF **207** associated with the sender and/or recipient. The symmetric keys may be generated based on one or more root keys provided by Root KDF **203** to Send KDF **205** and/or Receive KDF **207**. For example, Root KDF **203-1** may provide a root key output based on a public key and a private key to Send KDF **205-1**, which may generate a send key (e.g., based on the root key output and one or more private keys) which may be used to encrypt communications to be sent by messaging client **201-1**. The private key based on which the root key output is generated may be privately maintained by UE **101-1** (e.g., not sent to UE **101-2**), while the public key may be shared with UE **101-2** via private blockchain system **109**, in accordance with embodiments described herein. Root KDF **203-1** may provide a different root key output for each message sent and/or received by messaging client **201-1**, based on a public key exchange (e.g., via a Diffie-Hellman key exchange or other suitable procedure) with UE **101-2**, which may be performed each time a message is sent by UE **101-1** and/or UE **101-2**.

(33) For example, when UE **101-1** sends a message, Root KDF **203-1** may generate a root key output, and may further output a public key based on which the root key output was generated, to private blockchain system **109**. As discussed above, the public key may be provided with a session identifier (e.g., to identify a communication session between messaging client **201-1** and messaging client **201-2**), an identifier associated with UE **101-1**, an identifier associated with UE **101-2**, and/or other suitable information based on which UE **101-2** may identify that the public key recorded to private blockchain system **109** is associated with the communication session between UE **101-1** and UE **101-2**.

(34) UE **101-2** (e.g., Root KDF **203-2**) may obtain the root public key from private blockchain system **109** based on the session identifier or other suitable identifier, and may generate a root key output based on the obtained public key and a private key (e.g., a different private key than the private key used by Root KDF **203-1** to encrypt the communication). As noted above, using private blockchain system **109** to exchange public keys may maintain the integrity of the keys, such as by preventing malicious actors from modifying or “spoofing” keys. Further, using private blockchain system **109** may aid in situations where a given UE **101** (e.g., UE **101-2**, in this example) is “offline” or is otherwise unavailable to receive public keys from another UE **101** (e.g., UE **101-1**, in this example) at the time that the other UE **101** outputs the public keys.

(35) Root KDF **203-2** may provide the root key output to Receive KDF **207-2**, which may generate a receive key based on the root key output and one or more private keys. In accordance with the double ratchet techniques, Signal Protocol techniques, etc., the receive key generated by Receive KDF **207-2** may be the same as the send key output generated by Send KDF **205-1**. In this manner, these respective send and receive keys may be an identical symmetric key that may be used to encrypt and decrypt communications, such as the encrypted communication sent by messaging client **201-1** to messaging client **201-2** in this example.

(36) While an example is provided here in the context of UE **101-1** encrypting a communication, sending the encrypted communication to UE **101-2**, and UE **101-2** decrypting the communication, similar techniques may be performed (e.g., iteratively) for communications encrypted and sent by UE **101-2** to UE **101-1**, as denoted by the dashed lines in FIG. 2. For example, Root KDF **203-2** may generate a root key pair, output a public key of the root key pair to private blockchain system **109** for retrieval by UE **101-1**, generate a root key output based on the root key pair, and provide the root key output to Send KDF **205-2**. Send KDF **205-2** may generate a send key based on the root key output and one or more private keys, and messaging client **201-2** may encrypt a communication based on the send key. Further, Root KDF **203-1** may retrieve the public key from private blockchain system **109**, generate a root key output based on the retrieved public key and one or more private keys, and provide the root key output to Receive KDF **207-1**. Receive KDF **207-1** may generate a receive key based on the root key output and one or more private keys, and messaging client **201-1** may decrypt a communication received from messaging client **201-2**, as encrypted based on a matching send key.

(37) As noted above, private blockchain system **109** may be a “private” blockchain, in that only approved UEs **101** (or other suitable devices or systems) may participate in private blockchain system **109** as nodes **111** of private blockchain system **109**. In some embodiments, eligible participants may be identified based on an automated determination of contacts associated with one or more participant nodes **111** of private blockchain system **109**, and/or some other suitable technique. For example, as shown in FIG. 3, UE **101-1** may establish (at **301**) and/or join private blockchain system **109**. For example, UE **101-1** may establish a new private blockchain system **109** by generating a “genesis” block, and/or may join an existing private blockchain system **109**. In some embodiments, UE **101-1** may communicate with one or more devices or systems that facilitate the establishment of private blockchain system **109**.

(38) UE **101-1** may further automatically identify (at **303**) candidate participants (e.g., nodes **111** for private blockchain system **109**, devices or systems that are authorized to access information stored in private blockchain system **109**, or the like). For example, UE **101-1** may identify contacts in an address book, call history, and/or other type of contact list associated with UE **101-1**. Additionally, or alternatively, UE **101-1** and/or some other device or system may utilize artificial intelligence/machine learning (“AI/ML”) techniques or other suitable techniques to identify UEs **101** with which UE **101-1** has communicated.

(39) UE **101-1** may output (at **305**) an invitation to join (e.g., as a node **111** and/or as an authorized viewer of information stored in private blockchain system **109**) to identified UEs **101-2**, **101-3**, and **101-4**. In some embodiments, UE **101-1** (e.g., an application or application programming interface (“API”) associated with UE **101-1**) may communicate with contacts in a contact list associated with UE **101-1** (e.g., where the contacts include UEs **101-2**, **101-3**, and **101-4** in this example) via one or more suitable messaging or other types of communication applications, and may identify particular ones of the contacts participate in a key escrow or other type of key exchange. UE **101-1** may output an invitation to private blockchain system **109** to the identified contacts via the messaging or other communication applications and/or some other suitable communication pathway. In some embodiments, the invitation may include identifying information associated with UE **101-1**, such as an IP address associated with UE **101-1**, a Mobile Directory Number (“MDN”) associated with UE **101-1**, an International Mobile Station Equipment Identity (“IMEI”) associated

with UE **101-1**, an International Mobile Subscriber Identity (“IMSI”) associated with UE **101-1**, and/or some other suitable identifier.

(40) UEs **101-2**, **101-3**, and **101-4** may accordingly join (at **307**, **309**, and **311**, respectively) private blockchain system **109**. For example, such UEs **101-2**, **101-3**, and **101-4** may register as nodes **111** using information provided (at **305**) by UE **101-1**, and/or may access information stored in private blockchain system **109** using such information. In this manner, private blockchain system **109** may be “private” inasmuch as UEs **101** which were not invited (at **305**) to private blockchain system **109** may not access the information stored in private blockchain system **109**.

(41) FIG. **4** illustrates an example of the use of private blockchain system **109** to exchange public keys in the encryption and/or decryption of secure communications between UEs **101** participating in a secure communication session. In this example, UEs **101-1**, **101-2**, **101-3**, and **101-4** may be nodes **111** of private blockchain system **109**. Further, UEs **101-1** and **101-3** may be participants in a communication session (e.g., via respective messaging clients **201** associated with UEs **101-1** and **101-3**). For example, UE **101-1** and UE **101-3** may establish (at **401**) a secure communication session using any suitable session establishment technique or protocol. The communication session may be associated with a session identifier or other suitable mechanism by which UEs **101-1** and/or **101-3** may identify the communication session. For example, an initiator of the communication session may generate or otherwise determine an identifier for the communication session, and/or UEs **101-1** and **101-3** may otherwise negotiate or determine an identifier for the communication session.

(42) UE **101-1** may further generate (at **403**) an asymmetric key pair, which may be used as part of a double ratchet technique, a Signal Protocol technique, and/or some other encryption technique used to secure communications between UEs **101-1** and **101-3**. For example, the asymmetric key pair generated (at **403**) by UE **101-1** may include a public key used in the generation of a send key used to encrypt communications sent to UE **101-3**. While not explicitly shown in FIG. **4**, UE **101-3** may also generate one or more keys that may be used in the encryption and/or decryption of messages between UEs **101-1** and **101-3**, in accordance with the double ratchet technique, Signal Protocol technique, and/or other suitable encryption technique.

(43) UE **101-1** may further provide (at **405**) the public key of the generated asymmetric key pair to private blockchain system **109**. As noted above, UEs **101-1**, **101-2**, **101-3**, and **101-4** may be nodes **111** of private blockchain system **109**. In other examples, UE **101-3** (e.g., a participant in the secure communication session) may not be a node **111** of private blockchain system **109**, and thus UE **101-1** may not directly provide (at **405**) the public key to UE **101-3**. When providing (at **405**) the public key to private blockchain system **109**, UE **101-1** may include the session identifier of the communication session between UEs **101-1** and **101-3**, and/or some other suitable identifier based on which UE **101-3** may identify (at **407**) the public key provided by UE **101-1**.

(44) For example, as discussed above, UE **101-3** may retrieve (at **407**) the public key from private blockchain system **109** based on identifying a record in private blockchain system **109** that includes the session identifier or other suitable identifier associated with UEs **101-1** and/or **101-3**. In situations where multiple blocks exist with the session identifier, UE **101-3** may identify a latest (e.g., newest) block. For example, in accordance with the double ratchet and/or Signal Protocol techniques, a new public key may be provided each time a message is sent and/or received by UE **101-1** and/or UE **101-3**. As such, identifying a latest public key associated with the communication session between UEs **101-1** and **101-3** may facilitate the secure communications according to the double ratchet and/or Signal Protocol techniques.

(45) UE **101-1** may further encrypt (at **409**) a message for UE **101-3** using a symmetric key (e.g., a send key, as similarly discussed above) that UE **101-1** generates based on the public key, one or more private keys, and/or one or more KDFs (e.g., Send KDF **205**). UE **101-1** may proceed to output (at **411**) the encrypted message to UE **101-3**. UE **101-3** may decrypt (at **413**) the message using a symmetric key (e.g., the same symmetric key as UE **101-1** used to encrypt (at **409**) the

message), which may be generated by one or more KDFs (e.g., Receive KDF **207**) based on the retrieved (at **407**) public key.

(46) While FIG. **4** shows an example of one message being sent from UE **101-1** to UE **101-3**, similar techniques may be employed when UE **101-1** sends additional messages to UE **101-3**, and/or when UE **101-3** sends messages to UE **101-1**. For example, UE **101-3** may generate a subsequent asymmetric key pair that is derived from and/or is otherwise based on the retrieved (at **407**) public key, and may output a public key of the subsequent asymmetric key pair to private blockchain system **109**. UE **101-3** may utilize this subsequently generated public key to generate a send key, and UE **101-1** may utilize this subsequently generated public key, retrieved from private blockchain system **109**, to generate a receive key, as described above.

(47) FIGS. **5** and **6** illustrate example processes **500** and **600**, respectively, for utilizing blockchain techniques to share and/or exchange public keys. In the examples provided herein, the key exchange may be part of a Diffie-Hellman key exchange, a key exchange portion of a Signal Protocol technique, a key exchange portion of a double ratchet technique, and/or some other suitable secure exchange of public keys. In some embodiments, some or all of process **500** and/or **600** may be performed by one or more UEs **101** (e.g., UEs **101** engaged in a communication session with each other and/or one or more other UEs **101**). In some embodiments, one or more other devices may perform some or all of processes **500** and/or **600** in concert with, and/or in lieu of, UE **101**.

(48) As shown, process **500** may include establishing (at **502**) a communication session with one or more UEs **101**. The communication session may be associated with a session identifier or other identifier based on which participant UEs **101** of the communication session may identify messages exchanged with one another and/or public keys, associated with the communication session, recorded to private blockchain system **109** in accordance with embodiments described herein.

(49) Process **500** may further include generating (at **504**) a set of root keys, which may include an asymmetric key pair. For example, as discussed above, the root keys may include a private key which may be maintained securely by UE **101**, and a public key which may ultimately be shared with one or more other UEs **101** (e.g., another participant in the communication session).

(50) Process **500** may additionally include recording (at **506**) the public root key (e.g., the public key of the asymmetric key pair generated at **504**) to private blockchain system **109**. For example, UE **101** may output the public key to private blockchain system **109**, and private blockchain system **109** may propagate the information to one or more nodes **111** of private blockchain system **109**. Nodes **111** may form a consensus regarding the addition of the provided public key to one or more records of private blockchain system **109**, and may maintain an immutable record of the provided public key. In some embodiments, the record may include a timestamp, a block identifier, and/or other mechanism by which a recency or age of the record may be determined. As similarly described above, the record may further include a communication session identifier, an identifier of UE **101** from which the information was received, and/or other suitable information.

(51) Process **500** may also include utilizing (at **508**) the root keys to generate a send key. For example, as discussed above, Root KDF **203** may utilize the root keys (e.g., including the root public key, as well as the root private key and/or one or more other keys) to generate a root key output, and may provide the root key output to another KDF, such as Send KDF **205**. Send KDF **205** may generate a send key based on the root key output from Root KDF **203** and/or based on one or more other keys or functions. As discussed above, the generated send key may be a symmetric key that may be identical to a receive key generated by a recipient of the public key, based on a Signal Protocol technique, a double ratchet technique, etc.

(52) Process **500** may further include encrypting (at **510**) a communication using the send key. Process **500** may additionally include outputting (at **512**) the encrypted communication via the communication session, such as to a UE **101** with which the communication session was established (at **502**).

(53) As shown in FIG. 6, process **600** may include establishing (at **602**) a communication session. For example, as similarly discussed above, UE **101** may establish a communication session with one or more other UEs **101**. The communication session may be associated with a suitable identifier, based on which the UEs **101** participating in the communication session may identify communications associated with the communication session, including encrypted messages, public keys, and/or other information.

(54) Process **600** may further include receiving (at **604**) an encrypted communication via the communication session. For example, the communication may have been encrypted using one or more KDFs, where the input to the one or more KDFs include the public root key and one or more other keys (e.g., one or more private keys, one or more KDF outputs, etc.).

(55) Process **600** may further include retrieving (at **606**) a public root key from private blockchain system **109**. In some embodiments, private blockchain system **109** and/or one or more devices or systems communicatively coupled to private blockchain system **109** may “push” the public root key to private blockchain system **109**. For example, the one or more devices or systems may identify a particular block that includes an identifier of UE **101** (e.g., where the particular block includes the identifier of UE **101** and a public root key), may identify a particular block that includes an identifier of the communication session (e.g., where the particular block includes the identifier of the communication session and a public root key), etc. Additionally, or alternatively, UE **101** may “pull” the information from private blockchain system **109** based on a suitable identifier (e.g., an identifier of UE **101**, an identifier of the communication session, etc.).

(56) Process **600** may additionally include utilizing (at **608**) the public root key to generate a receive key. For example, as discussed above, UE **101** may utilize one or more KDFs, such as Root KDF **203** and/or Receive KDF **207**, to generate a receive key. The receive key may be a symmetric key with respect to a key used to encrypt a communication received by UE **101** via the communication session. For example, as discussed above, the receive key may be identical to a send key used by a sender of the communication, where the send key was also generated based on one or more KDFs and at least the public root key. For example, the sender and UE **101** may perform double ratchet techniques, Signal Protocol techniques, or the like, to respectively encrypt and decrypt the communication, without transmitting or sharing the symmetric key itself.

(57) Process **600** may further include decrypting (at **610**) the communication, received via the communication session, using the generated receive key. For example, UE **101** may apply one or more functions to decrypt the receive message using the receive key, which, as discussed above, may be the same as a send key used to encrypt the communication.

(58) Process **600** may additionally include generating (at **612**) a new set of root keys based on the previously retrieved public root key and one or more other root keys. For example, once UE **101** generates the receive key, the public root key (retrieved at **606**) may be discarded and/or not used for sending messages from UE **101**. In some situations, UE **101** may maintain the public root key and apply techniques described above to decrypt subsequent communications from the sender in situations where the sender sends multiple communications in a row without any intervening messages from UE **101**. The new set of root keys may be generated based on the receive key (generated at **608**), the public root key (retrieved at **606**), and/or one or more other suitable keys. UE **101** may then proceed to output the newly generated public root key to private blockchain system **109**, as similarly described above with respect to operation **506** of process **500**, may generate (e.g., similar to operation **508**) a new send key based on the newly generated public root key, etc.

(59) Embodiments described above are presented in the context of communication sessions between two UEs **101**. Embodiments described herein may further provide for secure blockchain-based key exchange mechanisms for group communication sessions that include more than two participants (e.g., between three or more UEs **101**). For example, techniques described below provide for a blockchain-based key exchange such that group messages may be encrypted such that

only participants in the group are able to access (e.g., decrypt) the messages. Further, as discussed below, situations may arise in which a participant in a communication session receives an encrypted message from another participant, but does not possess one or more keys needed to decrypt the message (e.g., a public key associated with the sender), and/or has not received confirmation from the blockchain that the participant is in possession of such keys. Techniques described below provide for the recipient to securely obtain the one or more keys from blockchain system **109** in order to decrypt the message. Further, embodiments described below provide for the use of one or more devices or systems, such as a Key Escrow System (“KES”), to serve as a fallback in instances where one or more keys cannot be obtained from blockchain system **109**, such as when a particular group message communication session participant does not participate in blockchain-based key exchange techniques described herein.

(60) FIG. 7 illustrates an example establishment of a group communication session that uses blockchain-based secure key exchange techniques. In this example, assume that the group communication session is between UEs **101-1**, **101-2**, and **101-3** (also referred to as “Participant_A,” “Participant_B,” and “Participant_C”). As shown, UE **101-1** may initiate (at **702**) a group communication session (also referred to as a “group chat”), such as by sending (at **702**) one or more messages to UEs **101-2** and **101-3**. For example, a messaging application (e.g., a particular messaging client **201**) of UE **101-1** may communicate with the same or a different messaging application (e.g., respective messaging clients **201**) of UEs **101-2** and **101-3**. The message(s) sent (at **702**) by UE **101-1** may include one or more Rich Communication Suite (“RCS”) messages or messages of another type or protocol. In some embodiments, the message(s) may include a session identifier associated with the group communication session, may include identifiers associated with UEs **101-1**, **101-2**, and/or **101-3** (e.g., IP addresses, SIP addresses, MDNs, etc.), and/or one or more other suitable identifiers. In some embodiments, the message(s) may include one or more instructions, indications, requests, etc. to implement blockchain-based key exchange for the communication session. In this example, assume that UEs **101-1**, **101-2**, and **101-3** support and/or have accepted the instruction, request, etc. to utilize blockchain-based key exchange.

(61) UEs **101-1**, **101-2**, and **101-3** may further select (at **704**) passive participants, or nodes **111**, for blockchain system **109** to securely maintain keys associated with the group communication session. For example, as similarly discussed above, UEs **101-1**, **101-2**, and **101-3** may automatically select contacts from respective address books, contacts lists, etc. In this example, UE **101-1** may select UE **101-4** as a node **111** for blockchain system **109**, UE **101-2** may select UE **101-5** as a node **111** for blockchain system **109**, and UE **101-3** may select UE **101-6** as a node **111** for blockchain system **109**.

(62) UEs **101-2** and **101-3** may indicate (at **706**) their respective selections of nodes **111** to UE **101-1**. For example, UE **101-2** may provide an identifier (e.g., MDN, IP address, etc.) of UE **101-5**, and UE **101-3** may provide an identifier of UE **101-6**. In some embodiments, UE **101-1** may have also provided (at **702**) a public key to UEs **101-2** and **101-3**. In some embodiments, UEs **101-2** and/or **101-3** may have provided (at **702**) respective public keys to UE **101-1** as part of the group communication session initiation. In some embodiments, UEs **101-2** and **101-3** may each encrypt (e.g., using suitable public keys, such as a public key of UE **101-1**) the messages sent (at **706**) to UE **101-1**, including the identifiers of UEs **101-5** and **101-6**.

(63) UE **101-1** may generate (at **708**) a full list of blockchain participants, including active participant UEs **101-1**, **101-2**, and **101-3**, as well as passive participant UEs **101-4**, **101-5**, and **101-6**. UE **101-1** may provide (at **710**) the full list of blockchain participants to UEs **101-2** and **101-3**, thus providing for UEs **101-2** and **101-3** to be aware of, and therefore able to communicate with, blockchain system **109** once established. In some embodiments, UE **101-1** may encrypt the messages (sent at **710**) with a private key, which may be associated with the public key provided at **702**. For example, the public and private keys may be associated with the same asymmetric key pair. UEs **101-2** and **101-3** may be able to use the public key (received at **702**) of UE **101-1** to

decrypt the provided list of blockchain participants. Additionally, or alternatively, UE **101-1** may encrypt the messages (sent at **710**) with public keys of respective recipients. For example, UE **101-1** may encrypt the message (sent at **710**) for UE **101-2** using a public key (e.g., as provided at **702**) of UE **101-2**, and may encrypted the message for UE **101-3** using a public key of UE **101-3**.

(64) UE **101-1** may proceed to establish (at **712**) blockchain system **109**, which may include generating a genesis block, providing the genesis block to nodes **111** of blockchain system **109** (e.g., UEs **101-2** through **101-6**), and/or other operations to establish blockchain system **109**. In some embodiments, the genesis block or other data included in or associated with blockchain system **109** may include or may be based on the session identifier associated with the group communication session. In this manner, different blockchain systems **109** may be associated with different communication sessions and may have different session identifiers.

(65) As shown in FIG. **8**, in situations where active participants of a group communication session (e.g., UEs **101-1**, **101-2**, and **101-3**) support and/or otherwise participate in blockchain-based key exchange techniques, such active participants may record suitable keys (e.g., public keys that may be used to derive symmetric keys used to encrypt and/or decrypt messages, as discussed above) to private blockchain system **109**, and may retrieve and/or validate suitable keys (e.g., keys associated with other active participants) from private blockchain system **109**. For example, when obtaining keys from blockchain system **109**, a particular UE **101** obtaining such keys may perform one or more cross-validation techniques in order to verify that the keys are accurate and/or have not been compromised in some way (e.g., a “man-in-the-middle” attack or some other type of malicious action).

(66) Situations may arise in which a participant of a group communication session does not support blockchain-based key exchange techniques, and/or otherwise does not implement or participate in the blockchain-based key exchange techniques described above. For example, a particular UE **101** may not have an API or application installed, via which UE **101** would serve as a node **111** or other type of participant in private blockchain system **109**. Additionally, or alternatively, a user of such UE **101** may not provide consent to utilize blockchain-based key exchange techniques, and/or may indicate a preference or instruction not to utilize blockchain-based key exchange techniques for one or more group communication sessions.

(67) For example, as shown in FIG. **9**, UE **101-1** may initiate (at **902**) a group communication session with UEs **101-2** and **101-3**, which may indicate outputting one or more RCS messages or other types of messages indicating the initiation of the group communication session. As similarly discussed above, UE **101-1** may indicate that the group communication session is associated with blockchain-based key exchange techniques. In this example, assume that UE **101-2** accepts the indication that the group communication session is associated with blockchain-based key exchange techniques, and/or does not deny the indication. For example, UE **101-2** may implement one or more APIs or applications via which UE **101-2** may serve as a node **111** of, and/or otherwise communicate with, private blockchain system **109**. As another example, a user of UE **101-2** may provide consent via a pop-up notification or other type of message or notification presented by UE **101-2** when receiving the group chat initiation from UE **101-1**. Accordingly, UE **101-2** may output a notification to UE **101-1** that UE **101-2** implements, and/or has received consent to implement, a blockchain-based key exchange for the initiated group communication session. Additionally, or alternatively, UE **101-2** may forgo outputting any such notification to UE **101-1**, based on which UE **101-1** may determine (e.g., based on the absence of such notification) that UE **101-2** will implement a blockchain-based key exchange for the initiated group communication session.

(68) On the other hand, UE **101-3** may output (at **904**) an indication that UE **101-3** has denied the blockchain-based key exchange for the initiated group communication session. Such indication from UE **101-3** may indicate, for example, that UE **101-3** does not support blockchain-based key exchange, that a user of UE **101-3** has not consented to blockchain-based key exchange for the initiated group communication session, and/or that UE **101-3** will otherwise not implement

blockchain-based key exchange for the group communication session. As one example, the group chat initiation message (at **902**) may include a RCS message or other type of message, that includes a flag, indicator, etc. associated with blockchain-based key exchange. UE **101-3** (e.g., a messaging application of UE **101-3** or other suitable type of application or API) may determine that the flag, indicator, etc. is not recognized and/or otherwise refers to one or more techniques (e.g., blockchain-based key exchange techniques) that are not implemented or supported by UE **101-3**. As another example, a user of UE **101-3** may have indicated a preference not to use blockchain-based key exchange for the group communication session, and/or has not indicated consent to use blockchain-based key exchange for the group communication session.

(69) As discussed herein, private blockchain system **109** may still be used for the group communication session, in addition to one or more other devices or systems (e.g., KES **901**), in order to provide for secure key exchange for the group communication session. The incorporation of KES **901** into the secure key exchange adds robustness as well as allows for a seamless experience between UEs **101** with different capabilities, preferences, and/or levels of user consent to use blockchain-based key exchange techniques.

(70) UE **101-1** may further perform (at **906**) one or more operations to establish private blockchain system **109**, as similarly discussed above. For example, UE **101-1** may communicate with UE **101-2** (which has not denied blockchain-based key exchange techniques) and/or one or more other UEs **101** (e.g., passive participants, as discussed above) to establish a particular private blockchain system **109** associated with the group communication session. Since UE **101-3** has denied (at **904**) the use of blockchain-based key exchange for the group communication session, the establishment (at **906**) of private blockchain system **109** may, in some embodiments, not include communicating with UE **101-3** to establish private blockchain system **109**.

(71) UEs **101-1**, **101-2**, and **101-3** may further share respective public keys to facilitate the group communication session. For example, as discussed above, UEs **101-1**, **101-2**, and **101-3** may share public keys so that such public keys may be used by other respective UEs **101-1**, **101-2**, and **101-3** of the group communication session to encrypt and/or decrypt messages (e.g., using symmetric keys, as discussed above, and/or in some other suitable manner). In some embodiments, such keys may be shared with an identifier (e.g., MDN, IP address, etc.) of a particular UE **101** with which the keys are associated, a session identifier associated with the group communication session, a key identifier that is unique to each key, and/or other suitable value.

(72) In accordance with some embodiments, UE **101-1** may share (at **908**) a public key associated with UE **101-1** (sometimes referred to in the figures as “Public key A” for the sake of clarity) to private blockchain system **109**. For example, since one or more UEs **101** of the group communication session (e.g., UEs **101-1** and **101-2**) participate in blockchain-based key exchange techniques, UE **101-1** may determine that the public key associated with UE **101-1** should be shared to private blockchain system **109**. Additionally, since one or more UEs **101** of the group communication session (e.g., UE **101-3**, in this example) do not participate in blockchain-based key exchange techniques, UE **101-1** may further determine that the public key associated with UE **101-1** should be shared to KES **901**. For example, based on receiving (at **904**) the indication from UE **101-3** that UE **101-3** has denied, does not support, etc. blockchain-based key exchange techniques for the group communication session, UE **101-1** may share its public key to KES **901** in addition to private blockchain system **109**.

(73) KES **901** may, for example, be a pre-designated device or system associated with a messaging application or API (e.g., messaging client **201**) via which UEs **101-1**, **101-2**, and **101-3** carry out the group communication session. Additionally, or alternatively, KES **901** may have been selected by UE **101-1**, elected by nodes **111** of private blockchain system **109**, selected by UE **101-3**, and/or selected in some other suitable manner. In some embodiments, as part of the group chat initiation (at **902**), and/or based on receiving (at **904**) a denial of UE **101-3** to participate in blockchain-based key exchange techniques, UE **101-1** may provide an IP address, URL, and/or other identifier of

KES **901** to other participants of the group communication session (e.g., UEs **101-2** and/or **101-3**). (74) In some embodiments, sharing (e.g., at **908**) a public key via KES **901** may include providing authentication and/or session information, such that public key may be obtained only by authorized entities. For example, when sharing the public key of UE **101-1** to KES **901**, UE **101-1** may provide a session identifier of the group communication session, and/or respective identifiers of UEs **101** associated with the group communication session (e.g., MDNs, IP addresses, etc. of UEs **101-1**, **101-2**, and/or **101-3**) with the public key. In this manner, when providing such keys, KES **901** may perform one or more authentication or authorization techniques in order to verify that an entity requesting public keys associated with the group communication session is authorized to receive such keys.

(75) In some embodiments, UE **101-2** may share (at **910**) a public key associated with UE **101-2** (e.g., Public key B) to private blockchain system **109**. For example, UE **101-2** may not be “aware” that UE **101-3** does not support and/or has denied the use of blockchain-based key exchange techniques. Further, since UE **101-3** does not support and/or has denied the use of blockchain-based key exchange techniques, UE **101-3** may share (at **912**) a public key associated with UE **101-3** (e.g., Public key C) to KES **901**.

(76) Thus, at this stage, private blockchain system **109** may maintain public keys associated with UEs **101-1** and **101-2**, and KES **901** may maintain public keys associated with UEs **101-1** and **101-3**. As discussed above, private blockchain system **109** and KES **901** may also maintain, with each key, a UE identifier, a group communication session identifier, a key identifier, and/or other suitable information based on which keys can be suitably identified and/or retrieved.

(77) As shown in FIG. **10**, UE **101-1** may have an outgoing group message **1001** to send to participants of the group communication session (i.e., UEs **101-2** and **101-3**, in this example). For example, a user of UE **101-1** may have provided input such as text, audio, an image, etc. to be sent to UEs **101-2** and **101-3** associated with the group communication session. Such input may be provided, for example, via a suitable messaging application executing at UE **101-1** (e.g., messaging client **201**). Based on receiving the input specifying outgoing message **1001**, UE **101-1** may determine that UE **101-1** does not have possession of public keys associated with the other active participants of the group communication session, namely public keys **107-2** and **107-3** associated with UEs **101-2** and **101-3** (e.g., Public key B and Public key C). Additionally, or alternatively, UE **101-1** may determine that UE **101-1** is in possession of public key **107-2** (e.g., may have received one or more public keys **107** by virtue of being a node **111** of blockchain system **109**), but may determine that such public key **107-2** has not been confirmed, validated, etc. by blockchain system **109**.

(78) Accordingly, as shown, UE **101-1** may obtain public key **107-2** from blockchain system **109**. As discussed above, for example, public key **107-2** may have previously been recorded to blockchain system **109** by UE **101-2**. In some embodiments, UE **101-1** may identify public key **107-2** by identifying a session identifier recorded with public key **107-2** and/or a UE identifier of UE **101-2** recorded with public key **107-2**, as discussed above. In some embodiments, UE **101-1** may identify public key **107-2** based on a key identifier associated with public key **107-2**. For example, based on detecting that UE **101-1** does not have possession of public key **107-2**, and/or based on detecting that UE **101-1** has received input specifying outgoing message **1001**, UE **101-1** may output requests to UEs **101-2** and **101-3** for key identifiers associated with public keys **107-2** and **107-3** of UEs **101-2** and **101-3**. In such embodiments, UEs **101-2** and **101-3** may provide such key identifiers, based on which UE **101-1** may search blockchain system **109** and/or otherwise identify corresponding public keys **107-2** and **107-3**.

(79) Since UE **101-1** has determined (e.g., at **904**) that UE **101-3** is not participating in blockchain-based key exchange techniques for the group communication session (and/or based on determining that UE **101-2** is participating in such blockchain-based key exchange techniques), UE **101-1** may obtain public key **107-2**, associated with UE **101-2**, from private blockchain system **109** (e.g., by

using one or more the identifiers discussed above). Further, UE **101-1** may obtain public key **107-3**, associated with UE **101-3**, from KES **901** (e.g., by using one or more the identifiers discussed above).

(80) That is, since UE **101-1** is “aware” that UE **101-3** is not participating in blockchain-based key exchange techniques, UE **101-1** may forgo attempting to retrieve public key **107-2** from private blockchain system **109**, as public key **107-2** is not expected to be located within private blockchain system **109**. In other words, UEs **101-1** may obtain public keys of blockchain-participating group communication session members (e.g., UE **101-2**, in this example) from private blockchain system **109**, and may obtain public keys of blockchain-nonparticipating group communication session members (e.g., UE **101-3**, in this example) from KES **901**.

(81) In some embodiments, when obtaining keys (e.g., public keys **107-2** and **107-3**, in this example) from blockchain system **109**, a particular UE **101** obtaining such keys (e.g., UE **101-1**, in this example) may perform one or more cross-validation techniques in order to verify that the keys are accurate and/or have not been compromised in some way (e.g., a “man-in-the-middle” attack or some other type of malicious action). For example, in this example, UE **101-1** may compute a hash of one or more blocks of blockchain system **109** from which public keys **107-2** and/or **107-3** were obtained, and/or may otherwise identify or generate information based on information recorded to blockchain system **109** that corresponds to public keys **107-2** and **107-3**.

(82) UE **101-1** may further communicate with one or more other nodes **111** of blockchain system **109** in order to verify the information (e.g., the hash of the one or more blocks from which public keys **107-2** and/or **107-3** were obtained). When quorum is reached (e.g., UE **101-1** has received at least a threshold quantity of confirmations from other nodes **111**) with respect to the hash or other suitable information, UE **101-1** may identify that the obtained public keys **107-2** and **107-3** are valid, accurate, etc., thus enhancing the security of the group communication session. Similar concepts may apply in situations where UE **101-1** already has possession of public keys **107-2** and **107-3** (e.g., in situations where UE **101-1** is a node **111** of blockchain system **109** and has therefore received public keys **107-2** and **107-3** when UEs **101-2** and **101-3** provided such public keys **107** to blockchain system **109**). For example, UE **101-1** may identify a local copy of blockchain system **109**, may obtain public keys **107-2** and **107-3** from the local copy of blockchain system **109**, and may perform a cross-validation process (e.g., may communicate with other nodes **111** of blockchain system **109**) to verify that public keys **107-2** and **107-3** from the local copy of blockchain system **109** is accurate, valid, etc.

(83) In other figures that discuss the obtaining of information (e.g., respective public keys **107**) from blockchain system **109**, a given UE **101** obtaining such information may perform similar operations in order to verify the authenticity, accuracy, etc. of the obtained information. Similarly, in other examples that discuss whether a given UE **101** is in possession of particular information (e.g., respective public keys **107**), such examples may also refer to whether the given UE **101** has received confirmations from other nodes **111** (e.g., at least a threshold quantity of nodes **111**, such that a quorum is reached) that such information is valid, even in instances where the given UE **101** already has possession of the information. That is, the given UE **101** may store a copy of one or more public keys **107** (e.g., within a local copy of blockchain system **109**), but may not be considered in “possession” of such public keys **107**, in accordance with some embodiments, if UE **101** has not received confirmation, validation, etc. from at least a threshold quantity of nodes **111** of blockchain system **109** that such copies of public keys **107** are valid, accurate, etc.

(84) As shown in FIG. **11**, after retrieving public keys **107-2** and **107-3** from blockchain system **109** and/or KES **901**, UE **101-1** may have possession of public keys **107-2** and **107-3**, as well as its own private key **105-1** (also referred to in the figures as “Private key A”) and public key **107-1**. As further shown, at this point, UEs **101-2** and **101-3** may have possession of their own respective private keys **105** and public keys **107**, but may not yet have possession of private keys **105** or public keys **107** associated with other UEs **101**.

(85) UE **101-1** may generate encrypted message package **1101** based on message **1001**, and send encrypted message package **1101** to UEs **101-2** and **101-3**. As shown, encrypted message package **1101** may include first portion **1103-1** and second portion **1103-2**. In some embodiments, encrypted message package **1101** may include other information, such as an identifier of UE **101-1**, a key identifier associated with public key **107-1**, and/or other suitable information.

(86) Portion **1103-1** may include a first encrypted version of message **1001**, while portion **1103-2** may include a second encrypted version of message **1001**. The first encrypted version of message **1001** (e.g., as included in portion **1103-1**) may, in some embodiments, be encrypted based on private key **105-1** (e.g., Private key A) and public key **107-2** (e.g., Public key B). For example, the first encrypted version of message **1001** may be encrypted based on a symmetric key derived from private key **105-1** and public key **107-2**. Similarly, the second encrypted version of message **1001** (e.g., as included in portion **1103-2**) may be encrypted based on private key **105-1** and public key **107-3** (e.g., a symmetric key derived from Private key A and Public key C).

(87) In some embodiments, portion **1103-1** may include an identifier of UE **101-2** and/or a key identifier associated with public key **107-2**, such that UE **101-2** is able to identify that UE **101-2** is the recipient of portion **1103-1** and/or that public key **107-2** has been used in the encryption of message **901** included in portion **1103-1**. Similarly, portion **1103-2** may include an identifier of UE **101-3**, a key identifier, and/or some other suitable information, such that UE **101-3** is able to identify that UE **101-3** is the recipient of portion **1103-2**.

(88) As shown in FIG. 12, UE **101-2** may retrieve public key **107-1** (e.g., Public key A) from blockchain system **109**, based on receiving encrypted message package **1101**. For example, UE **101-2** may identify that portion **1103-1** includes an encrypted message from UE **101-1** associated with the group communication session. For example, as discussed above, encrypted message package **1101** and/or respective portions **1103** may include a session identifier associated with the group communication session, key identifier, and/or identifiers of UEs **101-2** and/or **101-3**. As discussed above, retrieving public key **107-1** from blockchain system **109** may further include cross-validating the retrieved public key **107-1** with at least a threshold quantity of nodes **111** of blockchain system **109**.

(89) As further shown in FIG. 12, since UE **101-3** does not implement blockchain-based key exchange techniques, UE **101-3** may retrieve public key **107-1** from KES **901**. For example, UE **101-3** may identify that portion **1103-2** of encrypted message package **1101** includes an encrypted message from UE **101-1** associated with the group communication session. UE **101-3** may request public key **107-1** from KES **901**. For example, UE **101-3** may output a query, request, etc. that includes an identifier of UE **101-1**, an identifier of the group communication session, and/or other suitable information based on which public key **107-1** may be identified. In this example, since KES **901** has possession of public key **107-1**, and further assuming that KES **901** performs one or more suitable authentication and/or authorization operations to verify that UE **101-3** is authorized to obtain public key **107-1**, KES **901** may provide public key **107-1** to UE **101-3**.

(90) Once UEs **101-2** and **101-3** have obtained public key **107-1**, UE **101-2** may be in possession of public key **107-1** as well as its own private key **105-2** and public key **107-2**, and UE **101-3** may also be in possession of public key **107-1** as well as its own private key **105-3** and public key **107-3**. UEs **101-2** and **101-3** may accordingly decrypt their own respective portions **1103-1** and **1103-2** in order to extract original message **1001**. For example, as similarly discussed above, UE **101-2** may derive a symmetric key based on private key **105-2** (e.g., Private key B) and public key **107-1** (e.g., Public key A), and may use the symmetric key to decrypt portion **1103-1** in order to extract message **1001**. Similarly, UE **101-3** may derive a symmetric key based on private key **105-3** (e.g., Private key C) and public key **107-1** (e.g., Public key A), and may use the symmetric key to decrypt portion **1103-2** in order to extract message **1001**.

(91) As shown in FIG. 13, UE **101-3** may receive input specifying outgoing message **1301**, associated with the same group communication session. Based on operations described above, UE

101-1 may be in possession of keys **105-1**, **107-1**, **107-2**, and **107-3**; UE **101-2** may be in possession of keys **105-2**, **107-1**, and **107-2**; and UE **101-2** may be in possession of keys **105-3**, **107-1**, and **107-3**. Based on receiving the input specifying message **1301**, UE **101-3** may identify that UE **101-3** is not in possession of a public key associated with UE **101-2**. Accordingly, UE **101-3** may request public key **107-2**, associated with UE **101-2**, from KES **901**. That is, even though public key **107-2** has been recorded to private blockchain system **109**, UE **101-3** may not have access to private blockchain system **109**, may not implement one or more APIs or other interfaces associated with private blockchain system **109**, a user of UE **101-3** may have opted out of using private blockchain system **109**, etc. As such, UE **101-3** may instead request public key **107-2** from KES **901**.

(92) In some embodiments, based on receiving the request for public key **107-2**, KES **901** may output a request to UE **101-2** for public key **107-2**. For example, KES **901** may, in some situations, not possess public key **107-2**, and/or may possess an old or “stale” copy of public key **107-2**. For example, KES **901** may maintain public key **107-2**, but UE **101-2** may, in some embodiments, generate new public keys **107** on an ongoing basis, such as upon the generating or sending of new messages, based on the expiration of a timer, etc.

(93) Additionally, or alternatively, in some embodiments, even in situations where UE **101-3** does not record or submit information to private blockchain system **109**, UE **101-3** may have access to read, obtain, etc. information recorded to private blockchain system **109**. For example, during an establishment procedure of private blockchain system **109**, UE **101-1** may indicate, to nodes **111** of private blockchain system **109**, that UE **101-3** is authorized to retrieve information recorded to private blockchain system **109**. Additionally, or alternatively, private blockchain system **109** may be associated with one or more explorers, crawlers, etc. that maintain world state information associated with private blockchain system **109** (e.g., some or all information stored in private blockchain system **109**, such as up-to-date values of variables recorded to private blockchain system **109**). UE **101-3** may communicate with such explorers, crawlers, etc. in order to obtain public key **107-2**, in this example. In some embodiments, as discussed below, KES **901** may have access to read, obtain, etc. information recorded to private blockchain system **109**. In such embodiments, KES **901** may access private blockchain system **109** to obtain public key **107-2**, and may provide public key **107-2** to UE **101-3**.

(94) In some embodiments, as shown in FIG. **14**, KES **901** may request public key **107-2** from UE **101-2**, based on receiving the request from UE **101-3** for public key **107-2**. For example, in some embodiments, KES **901** may be isolated from private blockchain system **109** (e.g., may not have access to private blockchain system **109**, and/or may not be “aware” of private blockchain system **109**), and thus is incapable of retrieving public key **107-2** from private blockchain system **109**. UE **101-2** may accordingly provide the requested public key **107-2** to KES **901**, which may forward public key **107-2** to UE **101-3**. In this manner, UEs **101-2** and **101-3** need not communicate directly in order for UE **101-2** to provide public key **107-2** to UE **101-3**, thus reducing the possibility of public key **107-2** becoming exposed and the group communication session becoming compromised. Further, in this manner, private blockchain system **109** may be isolated from KES **901**, thus removing the reliance on trusting KES **901**, which may further enhance the security and integrity of the group communication session (e.g., at least from the standpoint of UEs **101-1** and **101-2**, which participate in private blockchain system **109**).

(95) Based on receiving (e.g., as discussed above with respect to FIG. **14**) public key **107-2**, and further based on the previous receiving (e.g., as discussed above with respect to FIG. **12**) of public key **107-1**, UE **101-3** may encrypt and send message **1301** to UEs **101-1** and **101-2**. For example, as shown in FIG. **15**, and as similarly discussed above, UE **101-3** may generate encrypted message package **1501**, which may include encrypted portions **1503-1** and **1503-2**. Encrypted portion **1503-1** may include, for example, a first encrypted version of message **1301**, and encrypted portion **1503-2** may include a second encrypted version of message **1301**. For example, the first encrypted

version of message **1301** may be encrypted based on a symmetric key associated with UEs **101-1** and **101-3** (e.g., Public key A and Private key C), and the second encrypted version of message **1301** may be encrypted based on a symmetric key associated with UEs **101-2** and **101-3** (e.g., Public key B and Private key C).

(96) As shown in FIG. 16, UEs **101-1** and **101-2** may receive encrypted message package **1501**. Since UE **101-1** has possession of public key **107-3** (e.g., based on operations discussed above), UE **101-1** may use previously obtained public key **107-3** (e.g., in conjunction with private key **105-1**) to decrypt portion **1503-1** of encrypted message package **1501**, in order to extract message **1301**. In this example, UE **101-2** does not yet have possession of public key **107-3**, and may therefore attempt to retrieve public key **107-3** from private blockchain system **109**.

(97) In this example, the attempted retrieval of public key **107-3** from private blockchain system **109** may be unsuccessful, as UE **101-3** has not provided public key **107-3** to private blockchain system **109** based on not participating in private blockchain system **109**. For example, UE **101-2** may determine that private blockchain system **109** does not include public key **107-3**, does not include any public keys **107** matching an identifier of UE **101-3**, does not include any public keys **107** matching an identifier of UE **101-3** and an identifier of the group communication session, and/or otherwise does not include public key **107-3** needed to decrypt portion **1503-2** of encrypted message package **1501**. In some embodiments, UE **101-2** may determine that the attempt to retrieve public key **107-3** was unsuccessful based on a “timeout” or expiration of a timer, indicating that UE **101-2** has searched private blockchain system **109** for at least a threshold amount of time (e.g., one second, ten seconds, one minute, etc.) without locating public key **107-3**.

(98) In other situations, the attempted retrieval of public key **107-3** from private blockchain system **109** may be unsuccessful for other reasons. For example, one or more nodes **111** of private blockchain system **109** may have failed, may be offline or unreachable, etc., and UE **101-2** may thus be unable to obtain a validated copy of public key **107-3**. For example, in some situations, a given key may have been receded to private blockchain system **109**. A given UE **101** may identify the key and may attempt to perform a cross-validation technique in which at least a threshold quantity (e.g., a quorum) of nodes **111** confirm the key as recorded to private blockchain system **109**. The cross-validation may fail if, for example, the threshold quantity of nodes **111** do not confirm, validate, etc. the key, which may occur if the threshold quantity of nodes **111** are offline or unreachable, or if the key has been compromised in some way.

(99) Thus, in situations where public key **107-3** is not available on private blockchain system **109** (and/or if cross-validation of public key **107-3** via private blockchain system **109** has failed), UE **101-2** may request public key **107-3** from KES **901**. In this example, KES **901** has possession of public key **107-3** based on previously receiving public key **107-3** (e.g., at **912**). In other examples, KES **901** may output a request to UE **101-3** for public key **107-3**, such as in instances where KES **901** does not have possession of public key **107-3**, instances in which KES **901** has possession of an “old” or “stale” copy of public key **107-3** (e.g., where public key **107-3** was received by KES **901** over a minute ago, over an hour ago, over a day ago, etc.), and/or in other suitable instances or embodiments. As such, KES **901** may serve as a fallback or measure of redundancy to further enhance the robustness of group messaging using blockchain-based key exchange techniques.

(100) In some embodiments, as noted above, KES **901** may have access to private blockchain system **109**, which may further enhance the functionality of KES **901**. For example, in some embodiments, KES **901** may record information to private blockchain system **109**, such as one or more keys (e.g., public key **107-3**, after receiving public key **107-3** from UE **101-3**). As shown in FIG. 17, UE **101-1** may initiate (at **1702**) a group communication session between UEs **101-1**, **101-2**, and **101-3**. As with the example of FIG. 9, assume that UE **101-2** supports and/or has accepted the blockchain-based key exchange techniques associated with the group communication session, and UE **101-3** may indicate (at **1704**) a denial of support and/or consent to use the blockchain-

based key exchange techniques.

(101) UE **101-1** may establish (at **1706**) private blockchain system **109** associated with the group communication session, which may include the selection of one or more nodes **111**, the creation and/or validation of a genesis block, the association of private blockchain system **109** with a session identifier associated with the group communication session, etc. As shown, establishing (at **1706**) private blockchain system **109** may include adding KES **901** as a node **111** of private blockchain system **109** and/or an authorized viewer of private blockchain system **109**. In some embodiments, KES **901** may be implemented by one or more UEs **101**. For example, a particular passive participant (e.g., UE **101-4**, **101-5**, and/or **101-6**, referring to the example of FIG. 7) may be designated as KES **901**. Additionally, or alternatively, KES **901** may be implemented by an application server, cloud system, datacenter, etc. In some embodiments, UE **101-1** may provide a session identifier, identifiers of group communication session participants (e.g., MDNs, IP addresses, etc. of UEs **101-1**, **101-2**, and **101-3**), and/or other suitable information to KES **901**.

(102) Based on being added as a node **111** and/or authorized viewer of private blockchain system **109**, KES **901** may be able to read and/or write to private blockchain system **109** (e.g., based on the particular permissions that were granted as part of the blockchain establishment procedure). As such, greater flexibility and robustness may be provided in the manner in which group communication session participants share and/or obtain keys.

(103) For example, as shown, UE **101-1** may share (at **1708**) public key **107-1** (e.g., Public key A) with private blockchain system **109**. As discussed above, public key **107-1** may be shared along with an identifier of UE **101-1**, a session identifier of the group communication session, etc. Based on being added to private blockchain system **109** (e.g., as a node **111**, an authorized viewer, etc.), KES **901** may “listen” to or monitor private blockchain system **109**, and may accordingly identify that public key **107-1** (shared at **1708**) is associated with the group communication session. As such, KES **901** may obtain (at **1710**) and/or cross-validate public key **107-1**.

(104) Similarly, UE **101-2** may share (at **1712**) public key **107-2** (e.g., Public key B) to private blockchain system **109**, and KES **901** may obtain (at **1714**) public key **107-2**. In this manner, KES **901** may retain local copies of public keys **107-1** and **107-2**, which have been cross-validated with nodes **111** of private blockchain system **109**. In this manner, in situations where private blockchain system **109** is unavailable or unreachable, cross-validated copies of public keys **107-1** and **107-2** may be obtainable from KES **901**, thus retaining the integrity provided by private blockchain system **109**. On the other hand, in situations where KES **901** is unavailable or unreachable, public keys **107-1** and **107-2** may still be available from private blockchain system **109**. As such, private blockchain system **109** and KES **901** may serve as fallbacks for each other, and the specific mode of retrieval (e.g., from private blockchain system **109** first with KES **901** as a fallback, or KES **901** first with private blockchain system **109** as a fallback) may be implemented based on factors or considerations such as security, amount of decentralization, trust (e.g., measure of trust of KES **901** and/or UEs **101** serving as nodes **111**), speed of retrieval and/or validation, etc.

(105) As further shown, UE **101-3** may share (at **1716**) public key **107-3** (e.g., Public key C) to KES **901**, as similarly described above. In some embodiments, KES **901** may share (at **1718**) public key **107-3** to private blockchain system **109**. In this manner, private blockchain system **109** may further maintain public key **107-3**, even though public key **107-3** has not been received by private blockchain system **109** directly from UE **101-3** (e.g., in situations in which UE **101-3** does not support or implement APIs, applications, etc. via which UE **101-3** may interact with private blockchain system **109**). In some embodiments, when sharing (at **1718**) public key **107-3** to private blockchain system **109**, KES **901** may include a digital signature that indicates the authenticity of public key **107-3** and/or of KES **901** (e.g., indicates that KES **901** is the entity that has recorded public key **107-3** to private blockchain system **109**), and/or includes some other mechanism that indicates that public key **107-3** is valid. In this manner, UEs **101-1** and **101-2** may be able to obtain public key **107-3** from private blockchain system **109**, without an initial attempt (e.g., as discussed

above with respect to FIG. 16) to obtain public key **107-3** from private blockchain system **109** and a subsequent request for public key **107-3** from KES **901**. In some embodiments, when obtaining public key **107-3** from private blockchain system **109** in this manner, UEs **101-1** and/or **101-2** may authenticate a signature of KES **901** and/or may otherwise validate public key **107-3**.

(106) FIG. **18** illustrates an example environment **1800**, in which one or more embodiments may be implemented. In some embodiments, environment **1800** may correspond to a Fifth Generation (“5G”) network, and/or may include elements of a 5G network. In some embodiments, environment **1800** may correspond to a 5G Non-Standalone (“NSA”) architecture, in which a 5G radio access technology (“RAT”) may be used in conjunction with one or more other RATs (e.g., a Long-Term Evolution (“LTE”) RAT), and/or in which elements of a 5G core network may be implemented by, may be communicatively coupled with, and/or may include elements of another type of core network (e.g., an evolved packet core (“EPC”)). As shown, environment **1800** may include UE **101**, RAN **1810** (which may include one or more Next Generation Node Bs (“gNBs”) **1811**), RAN **1812** (which may include one or more evolved Node Bs (“eNBs”) **1813**), and various network functions such as Access and Mobility Management Function (“AMF”) **1815**, Mobility Management Entity (“MME”) **1816**, Serving Gateway (“SGW”) **1817**, Session Management Function (“SMF”)/Packet Data Network (“PDN”) Gateway (“PGW”)-Control plane function (“PGW-C”) **1820**, Policy Control Function (“PCF”)/Policy Charging and Rules Function (“PCRF”) **1825**, Application Function (“AF”) **1830**, User Plane Function (“UPF”)/PGW-User plane function (“PGW-U”) **1835**, Unified Data Management function (“UDM”)/Home Subscriber Server (“HSS”) **1840**, Authentication Server Function (“AUSF”) **1845**, and KES **901**. Environment **1800** may also include one or more networks, such as Data Network (“DN”) **1850**.

(107) The example shown in FIG. **18** illustrates one instance of each network component or function (e.g., one instance of SMF/PGW-C **1820**, PCF/PCRF **1825**, UPF/PGW-U **1835**, UDM/HSS **1840**, and/or AUSF **1845**). In practice, environment **1800** may include multiple instances of such components or functions. For example, in some embodiments, environment **1800** may include multiple “slices” of a core network, where each slice includes a discrete set of network functions (e.g., one slice may include a first instance of SMF/PGW-C **1820**, PCF/PCRF **1825**, UPF/PGW-U **1835**, UDM/HSS **1840**, and/or AUSF **1845**, while another slice may include a second instance of SMF/PGW-C **1820**, PCF/PCRF **1825**, UPF/PGW-U **1835**, UDM/HSS **1840**, and/or AUSF **1845**). The different slices may provide differentiated levels of service, such as service in accordance with different Quality of Service (“QoS”) parameters.

(108) The quantity of devices and/or networks, illustrated in FIG. **18**, is provided for explanatory purposes only. In practice, environment **1800** may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. **18**. For example, while not shown, environment **1800** may include devices that facilitate or enable communication between various components shown in environment **1800**, such as routers, modems, gateways, switches, hubs, etc. Alternatively, or additionally, one or more of the devices of environment **1800** may perform one or more network functions described as being performed by another one or more of the devices of environment **1800**. Devices of environment **1800** may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of wired and wireless connections. In some implementations, one or more devices of environment **1800** may be physically integrated in, and/or may be physically attached to, one or more other devices of environment **1800**.

(109) UE **101** may include a computation and communication device, such as a wireless mobile communication device that is capable of communicating with RAN **1810**, RAN **1812**, and/or DN **1850**. UE **101** may be, or may include, a radiotelephone, a personal communications system (“PCS”) terminal (e.g., a device that combines a cellular radiotelephone with data processing and data communications capabilities), a personal digital assistant (“PDA”) (e.g., a device that may include a radiotelephone, a pager, Internet/intranet access, etc.), a smart phone, a laptop computer, a

tablet computer, a camera, a personal gaming system, an IoT device (e.g., a sensor, a smart home appliance, or the like), a wearable device, an Internet of Things (“IoT”) device, a Machine-to-Machine (“M2M”) device, or another type of mobile computation and communication device. UE **101** may send traffic to and/or receive traffic (e.g., user plane traffic) from DN **1850** via RAN **1810**, RAN **1812**, and/or UPF/PGW-U **1835**. In some embodiments, UE **101** may include, implement, etc. one or more nodes **111** of blockchain system **109** and/or KES **901**, as discussed above.

(110) RAN **1810** may be, or may include, a 5G RAN that includes one or more base stations (e.g., one or more gNBs **1811**), via which UE **101** may communicate with one or more other elements of environment **1800**. UE **101** may communicate with RAN **1810** via an air interface (e.g., as provided by gNB **1811**). For instance, RAN **1810** may receive traffic (e.g., voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **101** via the air interface, and may communicate the traffic to UPF/PGW-U **1835**, and/or one or more other devices or networks. Similarly, RAN **1810** may receive traffic intended for UE **101** (e.g., from UPF/PGW-U **1835**, AMF **1815**, and/or one or more other devices or networks) and may communicate the traffic to UE **101** via the air interface.

(111) RAN **1812** may be, or may include, a LTE RAN that includes one or more base stations (e.g., one or more eNBs **1813**), via which UE **101** may communicate with one or more other elements of environment **1800**. UE **101** may communicate with RAN **1812** via an air interface (e.g., as provided by eNB **1813**). For instance, RAN **1810** may receive traffic (e.g., voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **101** via the air interface, and may communicate the traffic to UPF/PGW-U **1835**, and/or one or more other devices or networks. Similarly, RAN **1810** may receive traffic intended for UE **101** (e.g., from UPF/PGW-U **1835**, SGW **1817**, and/or one or more other devices or networks) and may communicate the traffic to UE **101** via the air interface.

(112) AMF **1815** may include one or more devices, systems, Virtualized Network Functions (“VNFs”), etc., that perform operations to register UE **101** with the 5G network, to establish bearer channels associated with a session with UE **101**, to hand off UE **101** from the 5G network to another network, to hand off UE **101** from the other network to the 5G network, manage mobility of UE **101** between RANs **1810** and/or gNBs **1811**, and/or to perform other operations. In some embodiments, the 5G network may include multiple AMFs **1815**, which communicate with each other via the N14 interface (denoted in FIG. **18** by the line marked “N14” originating and terminating at AMF **1815**).

(113) MME **1816** may include one or more devices, systems, VNFs, etc., that perform operations to register UE **101** with the EPC, to establish bearer channels associated with a session with UE **101**, to hand off UE **101** from the EPC to another network, to hand off UE **101** from another network to the EPC, manage mobility of UE **101** between RANs **1812** and/or eNBs **1813**, and/or to perform other operations.

(114) SGW **1817** may include one or more devices, systems, VNFs, etc., that aggregate traffic received from one or more eNBs **1813** and send the aggregated traffic to an external network or device via UPF/PGW-U **1835**. Additionally, SGW **1817** may aggregate traffic received from one or more UPF/PGW-Us **1835** and may send the aggregated traffic to one or more eNBs **1813**. SGW **1817** may operate as an anchor for the user plane during inter-eNB handovers and as an anchor for mobility between different telecommunication networks or RANs (e.g., RANs **1810** and **1812**).

(115) SMF/PGW-C **1820** may include one or more devices, systems, VNFs, etc., that gather, process, store, and/or provide information in a manner described herein. SMF/PGW-C **1820** may, for example, facilitate the establishment of communication sessions on behalf of UE **101**. In some embodiments, the establishment of communications sessions may be performed in accordance with one or more policies provided by PCF/PCRF **1825**.

(116) PCF/PCRF **1825** may include one or more devices, systems, VNFs, etc., that aggregate

information to and from the 5G network and/or other sources. PCF/PCRF **1825** may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases and/or from one or more users (such as, for example, an administrator associated with PCF/PCRF **1825**).

(117) AF **1830** may include one or more devices, systems, VNFs, etc., that receive, store, and/or provide information that may be used in determining parameters (e.g., quality of service parameters, charging parameters, or the like) for certain applications.

(118) UPF/PGW-U **1835** may include one or more devices, systems, VNFs, etc., that receive, store, and/or provide data (e.g., user plane data). For example, UPF/PGW-U **1835** may receive user plane data (e.g., voice call traffic, data traffic, etc.), destined for UE **101**, from DN **1850**, and may forward the user plane data toward UE **101** (e.g., via RAN **1810**, SMF/PGW-C **1820**, and/or one or more other devices). In some embodiments, multiple UPFs **1835** may be deployed (e.g., in different geographical locations), and the delivery of content to UE **101** may be coordinated via the N9 interface (e.g., as denoted in FIG. **18** by the line marked “N9” originating and terminating at UPF/PGW-U **1835**). Similarly, UPF/PGW-U **1835** may receive traffic from UE **101** (e.g., via RAN **1810**, SMF/PGW-C **1820**, and/or one or more other devices), and may forward the traffic toward DN **1850**. In some embodiments, UPF/PGW-U **1835** may communicate (e.g., via the N4 interface) with SMF/PGW-C **1820**, regarding user plane data processed by UPF/PGW-U **1835**.

(119) UDM/HSS **1840** and AUSF **1845** may include one or more devices, systems, VNFs, etc., that manage, update, and/or store, in one or more memory devices associated with AUSF **1845** and/or UDM/HSS **1840**, profile information associated with a subscriber. AUSF **1845** and/or UDM/HSS **1840** may perform authentication, authorization, and/or accounting operations associated with the subscriber and/or a communication session with UE **101**.

(120) DN **1850** may include one or more wired and/or wireless networks. For example, DN **1850** may include an Internet Protocol (“IP”)-based PDN, a wide area network (“WAN”) such as the Internet, a private enterprise network, and/or one or more other networks. UE **101** may communicate, through DN **1850**, with data servers, other UEs **101**, and/or to other servers or applications that are coupled to DN **1850**. DN **1850** may be connected to one or more other networks, such as a public switched telephone network (“PSTN”), a public land mobile network (“PLMN”), and/or another network. DN **1850** may be connected to one or more devices, such as content providers, applications, web servers, and/or other devices, with which UE **101** may communicate.

(121) FIG. **19** illustrates an example Distributed Unit (“DU”) network **1900**, which may be included in and/or implemented by one or more RANs (e.g., RAN **1810**, RAN **1812**, or some other RAN). In some embodiments, a particular RAN may include one DU network **1900**. In some embodiments, a particular RAN may include multiple DU networks **1900**. In some embodiments, DU network **1900** may correspond to a particular gNB **1811** of a 5G RAN (e.g., RAN **1810**). In some embodiments, DU network **1900** may correspond to multiple gNBs **1811**. In some embodiments, DU network **1900** may correspond to one or more other types of base stations of one or more other types of RANs. As shown, DU network **1900** may include Centralized Unit (“CU”) **1905**, one or more Distributed Units (“DUs”) **1903-1** through **1903-N** (referred to individually as “DU **1903**,” or collectively as “DUs **1903**”), and one or more Radio Units (“RUs”) **1901-1** through **1901-M** (referred to individually as “RU **1901**,” or collectively as “RUs **1901**”).

(122) CU **1905** may communicate with a core of a wireless network (e.g., may communicate with one or more of the devices or systems described above with respect to FIG. **18**, such as AMF **1815** and/or UPF/PGW-U **1835**). In the uplink direction (e.g., for traffic from UEs **101** to a core network), CU **1905** may aggregate traffic from DUs **1903**, and forward the aggregated traffic to the core network. In some embodiments, CU **1905** may receive traffic according to a given protocol (e.g., Radio Link Control (“RLC”)) from DUs **1903**, and may perform higher-layer processing (e.g., may aggregate/process RLC packets and generate Packet Data Convergence Protocol

("PDCP") packets based on the RLC packets) on the traffic received from DUs **1903**.

(123) In accordance with some embodiments, CU **1905** may receive downlink traffic (e.g., traffic from the core network) for a particular UE **101**, and may determine which DU(s) **1903** should receive the downlink traffic. DU **1903** may include one or more devices that transmit traffic between a core network (e.g., via CU **1905**) and UE **101** (e.g., via a respective RU **1901**). DU **1903** may, for example, receive traffic from RU **1901** at a first layer (e.g., physical ("PHY") layer traffic, or lower PHY layer traffic), and may process/aggregate the traffic to a second layer (e.g., upper PHY and/or RLC). DU **1903** may receive traffic from CU **1905** at the second layer, may process the traffic to the first layer, and provide the processed traffic to a respective RU **1901** for transmission to UE **101**.

(124) RU **1901** may include hardware circuitry (e.g., one or more RF transceivers, antennas, radios, and/or other suitable hardware) to communicate wirelessly (e.g., via an RF interface) with one or more UEs **101**, one or more other DUs **1903** (e.g., via RUs **1901** associated with DUs **1903**), and/or any other suitable type of device. In the uplink direction, RU **1901** may receive traffic from UE **101** and/or another DU **1903** via the RF interface and may provide the traffic to DU **1903**. In the downlink direction, RU **1901** may receive traffic from DU **1903**, and may provide the traffic to UE **101** and/or another DU **1903**.

(125) RUs **1901** may, in some embodiments, be communicatively coupled to one or more Multi-Access/Mobile Edge Computing ("MEC") devices, referred to sometimes herein simply as "MECs" **1907**. For example, RU **1901-1** may be communicatively coupled to MEC **1907-1**, RU **1901-M** may be communicatively coupled to MEC **1907-M**, DU **1903-1** may be communicatively coupled to MEC **1907-2**, DU **1903-N** may be communicatively coupled to MEC **1907-N**, CU **1905** may be communicatively coupled to MEC **1907-3**, and so on. MECs **1907** may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE **101**, via a respective RU **1901**.

(126) For example, RU **1901-1** may route some traffic, from UE **101**, to MEC **1907-1** instead of to a core network (e.g., via DU **1903** and CU **1905**). MEC **1907-1** may process the traffic, perform one or more computations based on the received traffic, and may provide traffic to UE **101** via RU **1901-1**. In this manner, ultra-low latency services may be provided to UE **101**, as traffic does not need to traverse DU **1903**, CU **1905**, and an intervening backhaul network between DU network **1900** and the core network. In some embodiments, MEC **1907** may include, and/or may implement, some or all of the functionality described above with respect to one or more nodes **111** and/or KES **901**.

(127) FIG. **20** illustrates example components of device **2000**. One or more of the devices described above may include one or more devices **2000**. Device **2000** may include bus **2010**, processor **2020**, memory **2030**, input component **2040**, output component **2050**, and communication interface **2060**. In another implementation, device **2000** may include additional, fewer, different, or differently arranged components.

(128) Bus **2010** may include one or more communication paths that permit communication among the components of device **2000**. Processor **2020** may include a processor, microprocessor, or processing logic that may interpret and execute instructions. In some embodiments, processor **2020** may be or may include one or more hardware processors. Memory **2030** may include any type of dynamic storage device that may store information and instructions for execution by processor **2020**, and/or any type of non-volatile storage device that may store information for use by processor **2020**.

(129) Input component **2040** may include a mechanism that permits an operator to input information to device **2000** and/or other receives or detects input from a source external to **2040**, such as a touchpad, a touchscreen, a keyboard, a keypad, a button, a switch, a microphone or other audio input component, etc. In some embodiments, input component **2040** may include, or may be communicatively coupled to, one or more sensors, such as a motion sensor (e.g., which may be or

may include a gyroscope, accelerometer, or the like), a location sensor (e.g., a Global Positioning System (“GPS”)-based location sensor or some other suitable type of location sensor or location determination component), a thermometer, a barometer, and/or some other type of sensor. Output component **2050** may include a mechanism that outputs information to the operator, such as a display, a speaker, one or more light emitting diodes (“LEDs”), etc.

(130) Communication interface **2060** may include any transceiver-like mechanism that enables device **2000** to communicate with other devices and/or systems. For example, communication interface **2060** may include an Ethernet interface, an optical interface, a coaxial interface, or the like. Communication interface **2060** may include a wireless communication device, such as an infrared (“IR”) receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device **2000** may include more than one communication interface **2060**. For instance, device **2000** may include an optical interface and an Ethernet interface.

(131) Device **2000** may perform certain operations relating to one or more processes described above. Device **2000** may perform these operations in response to processor **2020** executing software instructions stored in a computer-readable medium, such as memory **2030**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **2030** from another computer-readable medium or from another device. The software instructions stored in memory **2030** may cause processor **2020** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(132) The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

(133) For example, while series of blocks and/or signals have been described above (e.g., with regard to FIGS. 1-17), the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel. Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

(134) The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

(135) In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

(136) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

(137) Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed by multiple devices. Further, multiple ones of the illustrated networks may be included in a single network, or a particular network may include multiple networks. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

(138) To the extent the aforementioned implementations collect, store, or employ personal information of individuals, groups or other entities, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various access control, encryption and anonymization techniques for particularly sensitive information.

(139) No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A first device, comprising: one or more hardware processors configured to: receive an encrypted communication from a second device, the encrypted communication being associated with a first key, wherein the encrypted communication is associated with a particular identifier; attempt to obtain, based on the particular identifier, the first key from a blockchain, wherein attempting to obtain the first key from the blockchain includes identifying a particular block of the blockchain that includes the particular identifier and that further specifies a second key; perform a cross-validation of the second key with at least a threshold quantity of nodes of the blockchain in order to validate the second key as being the first key; determine, based on performing the cross-validation, that at least the threshold quantity of nodes of the blockchain did not validate the second key as being the first key; determine, based on determining that the at least the threshold quantity of nodes of the blockchain did not validate the second key as being the first key, that the attempt to obtain the first key from the blockchain was unsuccessful; request, based on determining that the attempt to obtain the first key from the blockchain was unsuccessful, the first key from a third device, wherein the third device obtains the first key from the second device; receive the first key from the third device; and use the first key, as received from the third device, to decrypt the encrypted communication from the second device.

2. The first device of claim 1, wherein the encrypted communication is associated with a group communication session associated with the first device, the second device, and a fourth device, wherein the particular identifier is a first identifier, wherein the encrypted communication is a first encrypted communication, wherein the first device is further configured to: receive a second encrypted communication from the fourth device, the second encrypted communication being

associated with a third key; obtain the third key from the blockchain; and use the third key to decrypt the second encrypted communication from the fourth device.

3. The first device of claim 1, wherein the encrypted communication is associated with a group communication session associated with the first device, the second device, and a fourth device, wherein the fourth device is configured to establish the blockchain based on an initiation of a communication session associated with the first, second, and fourth devices, wherein establishing the blockchain includes indicating the establishment of the blockchain to the first and second devices, wherein the second device provides the first key to the third device based on receiving the indication of the establishment of the blockchain.

4. The first device of claim 3, wherein the second device provides the first key to the third device further based on at least one of: determining that the second device does not support recording information to the blockchain, or receiving an indication from a user of the second device not to join the blockchain.

5. The first device of claim 1, wherein the third device includes a Key Escrow System.

6. The first device of claim 1, wherein determining that at least the threshold quantity of nodes of the blockchain did not cross-validate the first key includes determining that at least the threshold quantity of nodes did not cross-validate the first key within a threshold amount of time.

7. The first device of claim 1, wherein the third device records the first key to the blockchain after receiving the first key from the second device.

8. A non-transitory computer-readable medium, storing a plurality of processor-executable instructions, wherein execution of the processor-executable instructions by a first device causes the first device to: receive an encrypted communication from a second device, the encrypted communication being associated with a first key, wherein the encrypted communication is associated with a particular identifier; attempt to obtain, based on the particular identifier, the first key from a blockchain, wherein attempting to obtain the first key from the blockchain includes identifying a particular block of the blockchain that includes the particular identifier and that further specifies a second key; perform a cross-validation of the second key with at least a threshold quantity of nodes of the blockchain in order to validate the second key as being the first key: determine, based on performing the cross-validation, that at least the threshold quantity of nodes of the blockchain did not validate the second key as being the first key; determine, based on determining that the at least the threshold quantity of nodes of the blockchain did not validate the second key as being the first key, that the attempt to obtain the first key from the blockchain was unsuccessful; request, based on determining that the attempt to obtain the first key from the blockchain was unsuccessful, the first key from a third device, wherein the third device obtains the first key from the second device; receive the first key from the third device; and use the first key, as received from the third device, to decrypt the encrypted communication from the second device.

9. The non-transitory computer-readable medium of claim 8, wherein the encrypted communication is associated with a group communication session associated with the first device, the second device, and a fourth device, wherein the particular identifier is a first identifier, wherein the encrypted communication is a first encrypted communication, wherein the plurality of processor-executable instructions further include processor-executable instructions to: receive a second encrypted communication from the fourth device, the second encrypted communication being associated with a third key; obtain the third key from the blockchain; and use the third key to decrypt the second encrypted communication from the fourth device.

10. The non-transitory computer-readable medium of claim 8, wherein the encrypted communication is associated with a group communication session associated with the first device, the second device, and a fourth device, wherein the fourth device is configured to establish the blockchain based on an initiation of a communication session associated with the first, second, and fourth devices, wherein establishing the blockchain includes indicating the establishment of the blockchain to the first and second devices, wherein the second device provides the first key to the

third device based on receiving the indication of the establishment of the blockchain.

11. The non-transitory computer-readable medium of claim 10, wherein the second device provides the first key to the third device further based on at least one of: determining that the second device does not support recording information to the blockchain, or receiving an indication from a user of the second device not to join the blockchain.

12. The non-transitory computer-readable medium of claim 8, wherein the third device includes a Key Escrow System.

13. The non-transitory computer-readable medium of claim 8, wherein determining that at least the threshold quantity of nodes of the blockchain did not cross-validate the first key includes determining that at least the threshold quantity of nodes did not cross-validate the first key within a threshold amount of time.

14. The non-transitory computer-readable medium of claim 8, wherein the third device records the first key to the blockchain after receiving the first key from the second device.

15. A method, comprising: receiving, by a first device, an encrypted communication from a second device, the encrypted communication being associated with a first key, wherein the encrypted communication is associated with a particular identifier; attempting to obtain, based on the particular identifier, the first key from a blockchain, wherein attempting to obtain the first key from the blockchain includes identifying a particular block of the blockchain that includes the particular identifier and that further specifies a second key; performing a cross-validation of the second key with at least a threshold quantity of nodes of the blockchain in order to validate the second key as being the first key; determining, based on performing the cross-validation, that at least the threshold quantity of nodes of the blockchain did not validate the second key as being the first key; determining, based on determining that the at least the threshold quantity of nodes of the blockchain did not validate the second key as being the first key, that the attempt to obtain the first key from the blockchain was unsuccessful; requesting, based on determining that the attempt to obtain the first key from the blockchain was unsuccessful, the first key from a third device, wherein the third device obtains the first key from the second device; receiving the first key from the third device; and using the first key, as received from the third device, to decrypt the encrypted communication from the second device.

16. The method of claim 15, wherein the encrypted communication is associated with a group communication session associated with the first device, the second device, and a fourth device, wherein the particular identifier is a first identifier, wherein the encrypted communication is a first encrypted communication, wherein the method further comprises: receiving a second encrypted communication from the fourth device, the second encrypted communication being associated with a third key; obtaining the third key from the blockchain; and using the third key to decrypt the second encrypted communication from the fourth device.

17. The method of claim 15, wherein the encrypted communication is associated with a group communication session associated with the first device, the second device, and a fourth device, wherein the fourth device is configured to establish the blockchain based on an initiation of a communication session associated with the first, second, and fourth devices, wherein establishing the blockchain includes indicating the establishment of the blockchain to the first and second devices, wherein the second device provides the first key to the third device based on receiving the indication of the establishment of the blockchain, and further based on at least one of: determining that the second device does not support recording information to the blockchain, or receiving an indication from a user of the second device not to join the blockchain.

18. The method of claim 15, wherein the third device includes a Key Escrow System.

19. The method of claim 15, wherein determining that at least the threshold quantity of nodes of the blockchain did not cross-validate the first key includes determining that at least the threshold quantity of nodes did not cross-validate the first key within a threshold amount of time.

20. The method of claim 15, wherein the third device records the first key to the blockchain after receiving the first key from the second device.
